

1. Basic GCD Properties

$$\gcd(a, 0) = a$$

$$\gcd(a, b) = \gcd(b, a \bmod b)$$

- Every common divisor of a and b divides $\gcd(a, b)$

2. Linear Combination Property

$$\gcd(a + m \cdot b, b) = \gcd(a, b) \quad \text{for any integer } m$$

3. Multiplicative Property

- If $\gcd(a_1, a_2) = 1$, then $\gcd(a_1 a_2, b) = \gcd(a_1, b) \cdot \gcd(a_2, b)$

4. GCD-LCM Product Identity

$$\gcd(a, b) \cdot \text{lcm}(a, b) = |a \cdot b|$$

5. GCD-LCM Distributive Laws

$$\gcd(a, \text{lcm}(b, c)) = \text{lcm}(\gcd(a, b), \gcd(a, c))$$

$$\text{lcm}(a, \gcd(b, c)) = \gcd(\text{lcm}(a, b), \text{lcm}(a, c))$$

6. Exponent GCD Property

$$\gcd(n^a - 1, n^b - 1) = n^{\gcd(a, b)} - 1$$

7. Totient Sum Representation

$$\gcd(a, b) = \sum_{\substack{k|a \\ k|b}} \phi(k)$$

8. Counting GCD Values

$$\sum_{i=1}^n [\gcd(i, n) = k] = \phi\left(\frac{n}{k}\right)$$

9. Sum of GCDs

$$\sum_{k=1}^n \gcd(k, n) = \sum_{d|n} d \cdot \phi\left(\frac{n}{d}\right)$$

10. Exponential GCD Sum

$$\sum_{k=1}^n x^{\gcd(k, n)} = \sum_{d|n} x^d \cdot \phi\left(\frac{n}{d}\right)$$

11. Reciprocal GCD Sum

$$\sum_{k=1}^n \frac{1}{\gcd(k, n)} = \frac{1}{n} \sum_{d|n} d \cdot \phi(d)$$

12. **Weighted Reciprocal GCD Sum**

$$\sum_{k=1}^n \frac{k}{\gcd(k, n)} = \frac{1}{2} \sum_{d|n} \phi(d)$$

13. **Modified GCD Sum**

$$\sum_{k=1}^n \frac{n}{\gcd(k, n)} = 2 \sum_{k=1}^n \frac{k}{\gcd(k, n)} - 1 \quad (n > 1)$$

14. **Coprime Pairs Count**

$$\sum_{i=1}^n \sum_{j=1}^n [\gcd(i, j) = 1] = \sum_{d=1}^n \mu(d) \left\lfloor \frac{n}{d} \right\rfloor^2$$

15. **Sum of GCD Pairs**

$$\sum_{i=1}^n \sum_{j=1}^n \gcd(i, j) = \sum_{d=1}^n \phi(d) \left\lfloor \frac{n}{d} \right\rfloor^2$$

16. **Coprime Pairs Product Sum**

$$\sum_{i=1}^n \sum_{j=1}^n i \cdot j [\gcd(i, j) = 1] = \sum_{i=1}^n \phi(i) i^2$$

17. **LCM Pairs Sum**

$$\sum_{i=1}^n \sum_{j=1}^n \text{lcm}(i, j) = \sum_{l=1}^n \left(\frac{(1 + \lfloor n/l \rfloor) \cdot \lfloor n/l \rfloor}{2} \right)^2 \sum_{d|l} \mu(d) l d$$

18. **Multiple GCD-LCM Relationship**

$$\begin{aligned} & \gcd(\text{lcm}(a, b), \text{lcm}(b, c), \text{lcm}(a, c)) \\ &= \text{lcm}(\gcd(a, b), \gcd(b, c), \gcd(a, c)) \end{aligned}$$

19. **Array GCD Property**

$$\gcd(A_L, A_{L+1}, \dots, A_R) = \gcd(A_L, A_{L+1} - A_L, \dots, A_R - A_{R-1})$$

20. **LCM Sum Formula**

- $\text{SUM} = \sum_{k=1}^n \text{lcm}(k, n)$

$$\text{SUM} = \frac{n}{2} \left(\sum_{d|n} \phi(d) \cdot d + 1 \right)$$